

$$\int x^2 \, \mathrm{d}x \tag{1}$$

$$\int_0^R \int_0^H \int_0^{2\pi} \rho z \phi \, \mathrm{d}\rho \, \mathrm{d}z \, \mathrm{d}\phi \tag{2}$$

$$\int_{[0,R]} \int_{[0,H]} \int_{[0,2\pi]} \rho z \phi \, \mathrm{d}\rho \, \mathrm{d}z \, \mathrm{d}\phi \tag{3}$$

$$\int_0^R \rho \, \mathrm{d}\rho \int_0^H z \, \mathrm{d}z \int_0^{2\pi} \phi \, \mathrm{d}\phi \tag{4}$$

$$\int_a^b f(x) \, \mathrm{d}x \tag{5}$$

$$\oint_{\mathcal{C}} f(\vec{r}) \cdot \mathrm{d}\vec{r} \tag{6}$$

$$\int \mathrm{d}x \tag{7}$$

$$\iiint\iiint_{\Omega} f(x_1,x_2,x_3,x_4,x_5) \, \mathrm{d}x_1 \, \mathrm{d}x_2 \, \mathrm{d}x_3 \, \mathrm{d}x_4 \, \mathrm{d}x_5 \tag{8}$$

$$\int_{x^2+y^2\leq 1} f(x,y) \, \mathrm{d}x \, \mathrm{d}y \tag{9}$$

$$\int_S xy \, \mathrm{d}x \, \mathrm{d}y \tag{10}$$

$$\int_{\mathbb{R}\times\mathbb{R}} xy \, \mathrm{d}x \, \mathrm{d}y \tag{11}$$

$$\oint_{\partial S} G(\vec{r}) \cdot \mathrm{d}\vec{r} \tag{12}$$

$$\int_0^R r^2 \, \mathrm{d}r \int_0^{2\pi} \sin \theta \, \mathrm{d}\theta \int_0^\pi \mathrm{d}\phi \tag{13}$$

$$\int r \cdot \mathrm{d}\vec{r} \tag{14}$$

$$\int 1 \, \mathrm{d}x + \int -\sin^2 x \, \mathrm{d}x \tag{15}$$

$$\int a \, \mathrm{d}a \tag{16}$$

$$\int b \cdot \mathrm{d}b \tag{17}$$